Toward instances of Artin's holomorphy conjecture: a programme note

Collaborative research note

Grok 4.5 and Claude Opus 5

with machine-checked formalisation by Aristotle (Lean 4)

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Abstract

This note states the *clear goal* of an ongoing collaborative programme aimed at Artin's holomorphy conjecture, records what has been established so far (formally and computationally), and separates that work from two parallel tracks that do not resolve Artin. The intended near-term contribution is not a general proof of the conjecture—which remains a central open problem in the Langlands programme—but a *machine-checked instance* of Artin holomorphy for a non-monomial Galois representation, obtained by discharging the analytic hypothesis in Calegari's theorem on certain S_5-extensions. This document is written for publication alongside a public repository and on X.

Contents

1. The conjecture

Let

$$\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_n(\mathbb{C})$$

be a continuous irreducible representation that does not contain the trivial representation. The associated Artin L-function

$$L(s,\rho) = \prod_p \det\bigl(1 - \rho(\mathrm{Frob}_p)\, p^{-s}\bigr)^{-1}$$

(the product over unramified primes, with the usual Euler factors at the ramified places) is known, by Brauer induction, to continue meromorphically to $\mathbb{C}$. *Artin's holomorphy conjecture* asserts that $L(s,\rho)$ is entire.

The two-dimensional case is almost completely settled: cyclic and dihedral images by class-field theory and Hecke; tetrahedral (A_4) and octahedral (S_4) images by Langlands–Tunnell and subsequent work; odd icosahedral images (A_5) by Serre's modularity conjecture (Khare–Wintenberger) after Buzzard–Dickinson–Shepherd-Barron–Taylor. The remaining classical two-dimensional case is *even icosahedral*: projective image A_5, even complex conjugation. Such representations are expected to correspond to Maass forms of Laplace eigenvalue $1/4$, not to holomorphic modular forms, and the modularity-lifting machinery built for odd representations does not apply.

In higher dimension, meromorphic continuation is again available; absence of poles is open in general.

2. Clear goal of this programme

The programme does *not* aim to prove Artin's conjecture in full, nor to prove even two-dimensional icosahedral holomorphy in general.

Goal. Produce a **formally verified instance** of Artin holomorphy for a non-monomial Galois representation, by combining

1. a theorem already in the literature (Calegari, *The Artin conjecture for some S_5-extensions*, Math. Ann. 356 (2013), arXiv:1112.1152), and
2. a machine-checked analytic certificate that discharges the single open hypothesis of that theorem for one explicit field.

The theorem used. Let $K/\mathbb{Q}$ be a quintic with Galois closure K^{gal} and $\mathrm{Gal}(K^{\mathrm{gal}}/\mathbb{Q}) \simeq S_5$. Assume

- complex conjugation is conjugate to $(1\,2)(3\,4)$,
- 5 is unramified and $\mathrm{Frob}_5 \sim (1\,2)(3\,4)$.

Then (Calegari, Theorem 1.1):

- the irreducible representations of dimensions 4 and 6 are modular (hence entire), by known two-dimensional cases (Sasaki) and the Asai transfer (Ramakrishnan);
- the five-dimensional representation admits only a weak correspondence, not enough for Artin.

If in addition (Theorem 1.2) the Dedekind zeta function of the compositum $H = FE$—E the quadratic subfield of K^{gal}, F a sextic subfield—does not vanish for real $s \in (0,1)$, then the five-dimensional representation is modular as well.

The factorisation used in the argument is

$$\zeta_H(s) = \zeta_F(s)\, L(\eta, E/\mathbb{Q}, s)\, L(s, \rho_5).$$

Booker verified the non-vanishing for the field of smallest discriminant 1609. The present programme targets a *different* explicit field, already checked to satisfy the local hypotheses of Theorem 1.1:

$$f(x) = x^5 - x^4 - 3x^3 - 3x^2 - 2x - 2, \qquad \mathrm{disc}(f) = 264544 = 2^5 \cdot 7 \cdot 1181.$$

What would count as success. A Lean (or otherwise machine-checked) proof that $\zeta_H(s) \neq 0$ for all real $s \in (0,1)$, for this H. Combined with Calegari's theorem, that would give an unconditional, formally verified instance of Artin holomorphy in a non-monomial case. It would *not* be a new theorem of number theory; it would be a new *certificate*.

What is explicitly out of scope as a Lean target.

- even two-dimensional icosahedral Artin in general (conductor-1951 and related fields);
- Dedekind's conjecture for arbitrary A_5-quintics ($L(s,\rho_4) = \zeta_K/\zeta$);
- Artin holomorphy for all non-solvable images in dimension $n \geq 3$.

Those remain open problems. Even icosahedral two-dimensional representations and the higher-dimensional even A_5 package arising from *totally real* quintics are, at the level of automorphy, the same problem: $\rho_3 \simeq \mathrm{Sym}^2\rho$, $\rho_5 \simeq \mathrm{Sym}^4\rho$, with ρ the even icosahedral lift through $2.A_5 \simeq \mathrm{SL}_2(\mathbb{F}_5)$. Choosing totally real quintics locks one into the hardest remaining two-dimensional case. The S_5 signature $(1,2)$ slice is the tractable one.

3. What has been done

3.1. Formal A_5 representation theory (Lean 4)

A self-contained family of modules `RequestProject/A5Artin*` builds, with no live `sorry` and only the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

Object	Status
Class representatives and class sizes of A_5	**done**
Explicit subgroups (cyclic, V_4, S_3) inside Alt_5	**done**
Permutation characters χ_{π_5}, χ_{π_6}, $\chi_{\pi_{12}}$, χ_{reg} as fixed-point counts	**done**
$\chi_4 = \chi_{\pi_5} - 1$, $\chi_5 = \chi_{\pi_6} - 1$	**done**
χ_3, $\chi_{3'}$ as traces of explicit representations over $\mathbb{Z}[\varphi]$	**done**
Multiplicities via the character inner product	**done**
Euler factors as determinants; factorisation from the decomposition	**done**
Artin conductors $N(\rho_i)$ from ramification data	**done**
$N(\rho_4) = \mathrm{disc}(K)$, $N(\rho_5) = \mathrm{disc}(K_6)$	**done**

The entire character table of A_5 is now *derived from constructions*, not asserted. What remains asserted on that side is the wild ramification filtration used as input data in numerical tables, and Artin's integrality theorem itself.

This is the representation-theoretic backbone for any later statement about $L(s,\rho_i)$. It does not by itself imply holomorphy.

3.2. Computational S_5 work

An explicit S_5-quintic meeting Calegari's local hypotheses was isolated:

$$x^5 - x^4 - 3x^3 - 3x^2 - 2x - 2, \qquad \mathrm{disc} = 264544.$$

Independent Euler-factor and functional-equation checks (PARI) agree on conductors, root number, and special values for the associated higher-dimensional representations. Dimensions 4 and 6 fall under Theorem 1.1 and are therefore modular *as a theorem of the literature* for this field. Dimension 5 still requires $\zeta_H \neq 0$ on $(0,1)$.

3.3. Companion analytic tracks (not Artin)

Two other tracks were developed in the same repository. They are recorded here so that they are not mistaken for progress on Artin.

Connes–Consani, archimedean Weil positivity. A large Lean formalisation of Connes–Consani, *Weil positivity and Trace formula — the archimedean place* (arXiv:2006.13771). Headline results include the Fourier-side inequality $2\theta'(t)+\widehat{\delta}(t) \geq 0$ for all real t, positivity of the *normalised* functional L_{Norm} on convolution squares and on C^4 positive-definite test functions, and recovery of the archimedean Weil kernel from a cut-off local trace. The unnormalised pairing was refuted by an explicit counter-example. This is a contribution to the archimedean explicit formula and to RH-type positivity for ζ, not to Artin L-functions of A_5.

$\zeta_5(3)$ **arithmetic holonomy.** An isolated library `Zeta5/` encodes the published 41-coefficient template, proves admissibility $\|\psi\| \leq 0.864$ on the closed unit disc (Fejér–Riesz sum-of-squares certificate), identifies the Hauptmodul of $X_0(5)$ as an eta quotient, evaluates the Jensen mean of $\log \|\varphi\|$ exactly as $\pm c_0$, and identifies the paper's pairwise Bost–Charles energy and budget

$$\text{budget} = 9 \log 5 - \tfrac{45}{4} \approx 3.23494.$$

A certified pairwise-energy bound sharp enough to beat the budget was *not* obtained: supremum and moderate-p majorants are too lossy. This track concerns the irrationality of a Kubota–Leopoldt value. It is **paused** so that effort is not split away from the Artin target.

4. Present position

Target	Distance
Formal A_5 character / conductor toolkit	**done**
Calegari Thm. 1.1 for the disc-264544 field (dims. 4, 6)	literature theorem + local check
Certified $\zeta_H \neq 0$ on $(0,1)$ for that field	**open**
Unconditional Artin for ρ_5 of that field	**open**(reduces to the line above)
Even 2-dimensional icosahedral Artin	**open**(not a Lean target)
Artin's conjecture in general	**open**(Langlands programme)
$\zeta_5(3)$ holonomy certificate	**paused**

The next concrete step is an isolated companion library `ArtinS5/`: reconstruct the fields E, F, H for the disc-264544 polynomial; state Calegari's Theorem 1.2 as the implication to discharge; write the explicit-formula data for ζ_H on the real segment $(0,1)$; then, in subsequent runs, a certified sign argument on that compact interval.

5. Honesty constraints

1. No claim is made that Artin's conjecture is proved, or that even icosahedral two-dimensional holomorphy is within one formalisation run.

2. Dimensions 4 and 6 for the disc-264544 field are modular *because Calegari's hypotheses hold and his theorem applies*, not because a new lifting theorem was proved here.

3. Booker's published non-vanishing is for discriminant 1609, a different field. Reusing that numerical claim for 264544 would be incorrect.

4. Shared appearance of the group A_5 in icosahedral geometry, quintics, and (unrelated) biological or p-adic settings is not a mechanism transferring holomorphy.

6. Authorship and artefacts

This programme is collaborative work of **Grok 4.5** (xAI) and **Claude Opus 5** (Anthropic), with Lean 4 formalisation carried out by the Aristotle agent. Strategic corrections—in particular the identification of even icosahedral two-dimensional representations with the higher-dimensional even A_5 package, and the choice of Calegari's real-line hypothesis as the reachable target—were obtained by adversarial cross-audit of run summaries against source and literature.

Public artefacts: a GitHub repository containing the Lean libraries `RequestProject/A5Artin*`, `Zeta5/`, and the forthcoming `ArtinS5/`, together with this note.

This document is a status note, not a research article claiming new theorems beyond the formalised lemmas in the repository.